

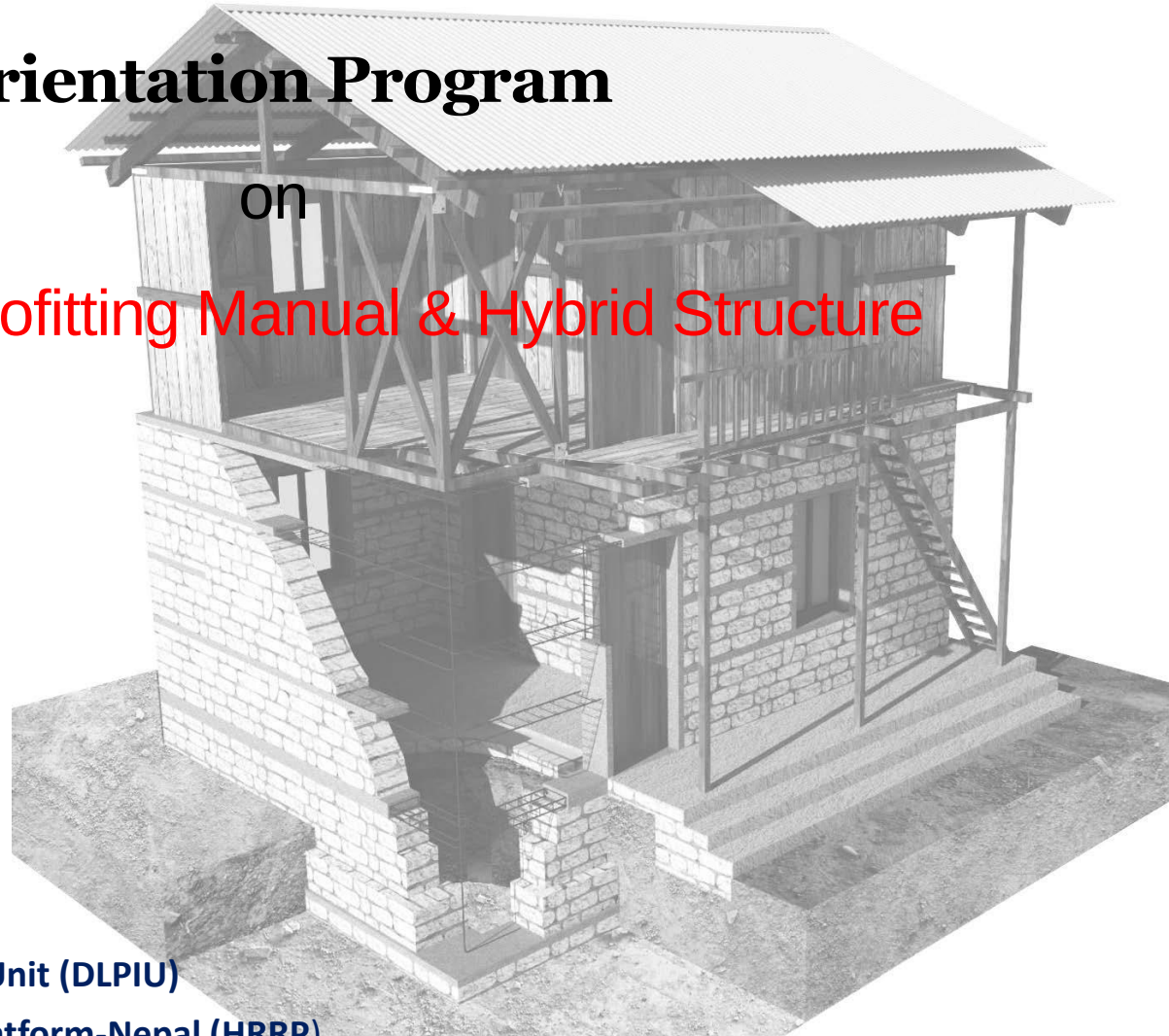


Housing Recovery and Reconstruction Platform-Nepal (HRRP)

Orientation Program

on

Repair & Retrofitting Manual & Hybrid Structure



Organized by:

National Reconstruction Authority(NRA)

MOUD/District Project Implementation Unit (DLPIU)

Housing Recovery and Reconstruction Platform-Nepal (HRRP)

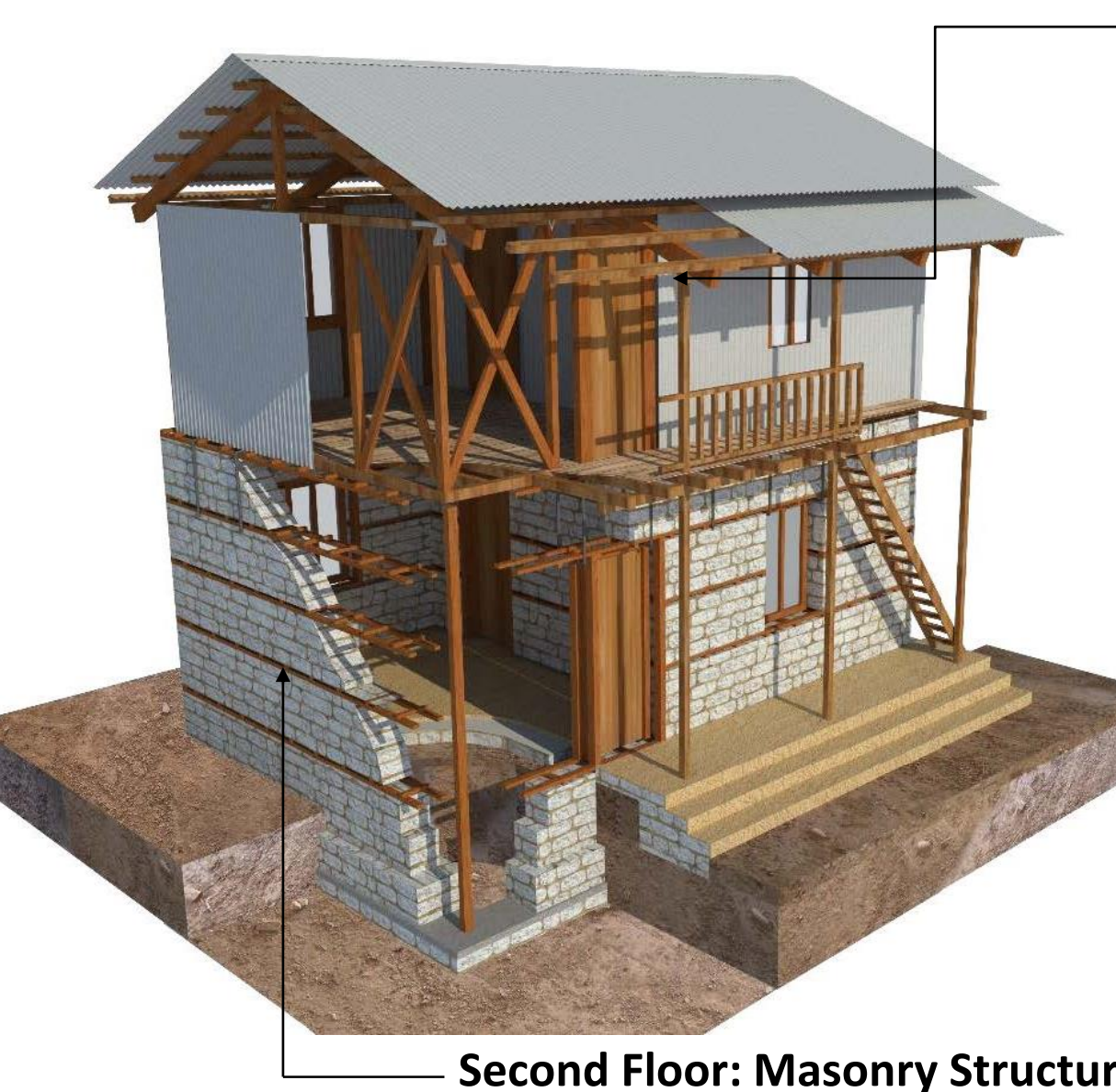
1. Background

- Low Strength Masonry with two or three storeys dominate Structures in rural areas
- Two or more than two storeys with mud mortar are common
- The MRs restrict the use of masonry above the ground floor Plus attic in RCC buildings
- The people tends to construct two or three storeys for living functional requirement

Therefore, in order to ensure the safety of these building against earthquake load and wind load, it has become an urgent task to make the construction guideline along with proper connection details and standards of hybrid structure

So, Hybrid structure is huge demand in the reconstruction field

What is Hybrid Structure ?



First floor: Frame Structure

Second Floor: Masonry Structure

Hybrid structure(Mix structure) is the combination of two or more type of structural system that is generally constructed with different technology and materials in accordance with level of floor.

Limitation

- Under the GoN housing reconstruction programme, this manual covers only hybrid structure that are newly constructed or under construction.
- It covers Residential building d fall under category 'C' and 'D' of NBC.



Only Covers Load bearing masonry structure in ground floor & frame structure with light weight wall in first floor.

Construction Sequence & Construction Methodology



1



Install Wooden base plate

2



Install vertical Member

3



Install Bracing

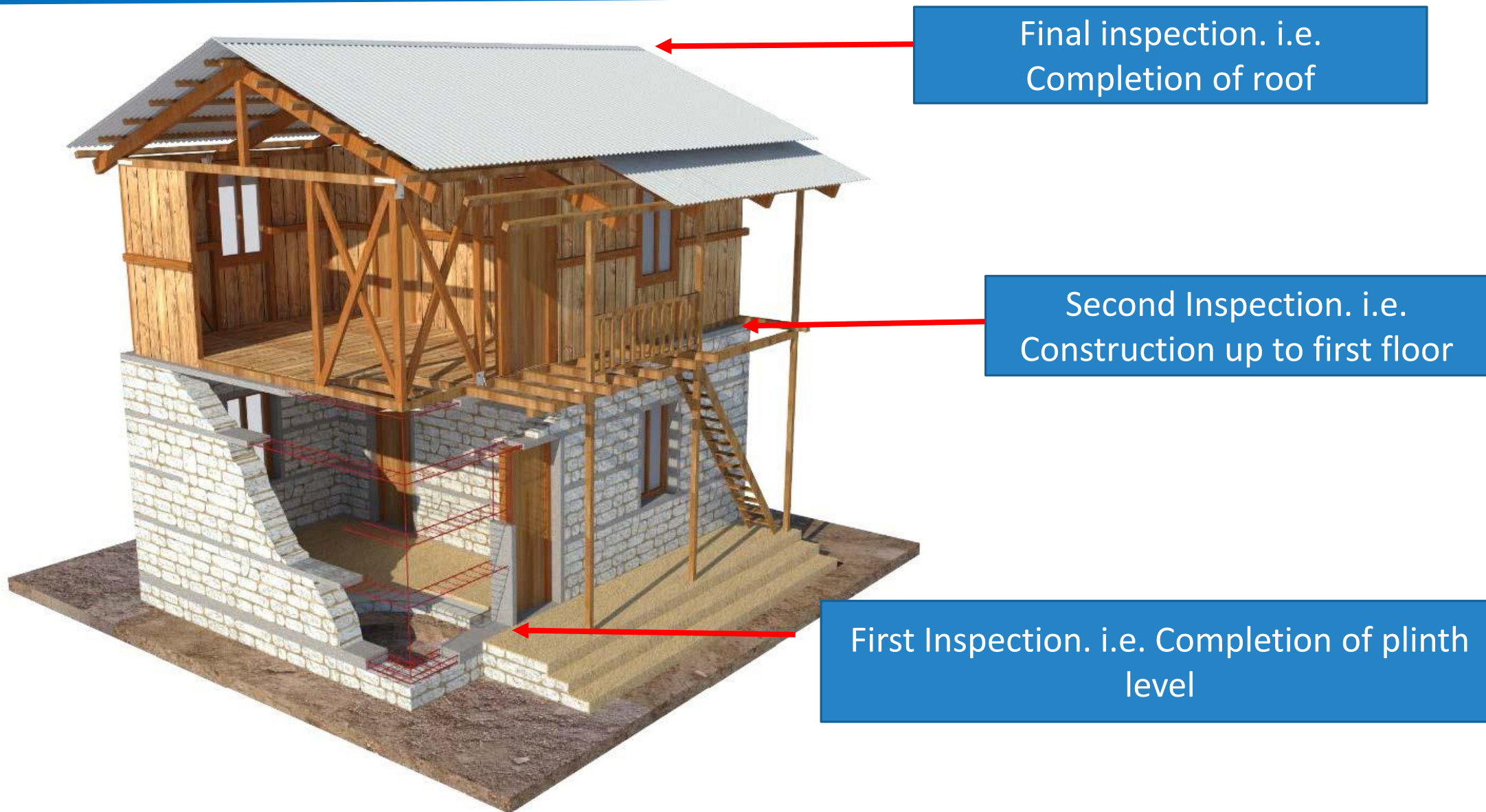
4



Construct wooden truss

The Ground floor is same as Load bearing Masonry Structures as guided by MRs .

Inspection Process



Technical Specification of Hybrid Structure

Key Inspection Items



1. Shape and Size of building
2. Materials
3. Connections and Joints
4. Frame action
5. Roof

1. Shape and Size of building

- No. of Story: Not more than 2 story
 - Height of wall: 1st floor not more than 2.5m
- Shape & Size: The wall line of upper storey shall be on the wall line of lower storey.



2. Materials

Requirements

No.	Category	Sub-category	Description
2.	Materials	Nail	Common wire nails shall be made of mild steel having a minimum tensile strength of 550N/mm ² . Nails with appropriate diameter and length shall be provided.
		Bolt	It shall be used in such the number, diameter, length, spacing as each specification.
		Metal Plate	It shall be used in such the number, diameter, length, spacing as each specification.
		Rebar	High strength deformed bars with $f_y = 415 \text{ Mpa} / 500 \text{ Mpa}$.
		Timber	Treated and well seasoned hard wood or locally available wood without knots shall be used..

Why important?

- Inadequate materials does not have sufficient stability and strength to withstand the lateral forces. Hence, use of these substandard materials might leads to the failure or ultimately collapse of the overall structure.
- Moisture can cause wooden surfaces to swell and deform. Excessive moisture will lead the wood to decay, caused by decay fungi that ruin the material completely.
- Shrinkage of wood on drying is relatively large. Joint loosen easily due to contraction in the direction perpendicular to fibers. Therefore dry wood shall be used with moisture content less than 20 %.
- Wood can decay from repeated change of moistures. Therefore seasoned wood should be used in construction.

Inspection methodology

It can be checked by the observation and measurement.

Technical specification

2.1 Nail

- The things that need to be checked in nail are as follows:

a) Nails diameter:

Nail diameter shall be in between $1/11$ and $1/6$ of the least thickest of the members to be connected.

b) Nails length:

The length of a nail shall be at least 2.5 times the thickness of the thinnest member and it shall penetrate the thicker member by 1.5 times the thickness of the thinner member, whichever is further.

c) Number of nails:

The number of nails in a group should not exceed 10 in one rows in the direction of the force.



2.2 Bolts:

- When a number of bolts are used in a joint, the allowable load in withdrawal or lateral resistance shall be the sum of the allowable loads for the individual bolts.



2.3 Metal Plate:

- The bolts shall be arranged in such the size, thickness, spacing as each design/specification.



2.4 Rebar:

- Heavily rusted rebar should not be used.
- After rubbing the steel bar, if stain is present on fingers, but if the flakes doesn't come off then the rust is acceptable.
- The thickness of rebar is checked by using vernier caliper. Ductility of rebar can be checked by bending it at 90° and if small cracks are found ductility is insufficient.



3. Connections and Joints

Requirements

No.	Category	Sub-category	Description
3.	Connections and joints	Connections between lower and upper structure	It shall be properly connected as per the specification.
		Joints of structural member	All the structural members shall be properly connected by nails, bolts and metal plate as per the specification.

Why important ?

Connections between lower and upper structure

- If there is poor connection between the ground floor and first floor with rigid structure, the building might tends to uplifting/rocking or sliding behavior, when the lateral load is imposed on to the structure.
- Bonding strength of stone masonry in mud mortar with wooden band is very poor, therefore, failure pattern mentioned above will occur easily.



Uplifting/Rocking



Sliding

Joints of structural member

- The failure of the joints connecting structural member such as vertical, horizontal and bracing frequently occurs. Structural member should be uniform, so that the frame will acts as earthquake resistance elements.

Technical specification

Connections between lower and upper structure

In case of floor band is made by **Reinforcement Concrete**



Construction methods are from

P34

In case of stone masonry in **mud mortar** with **wooden band**



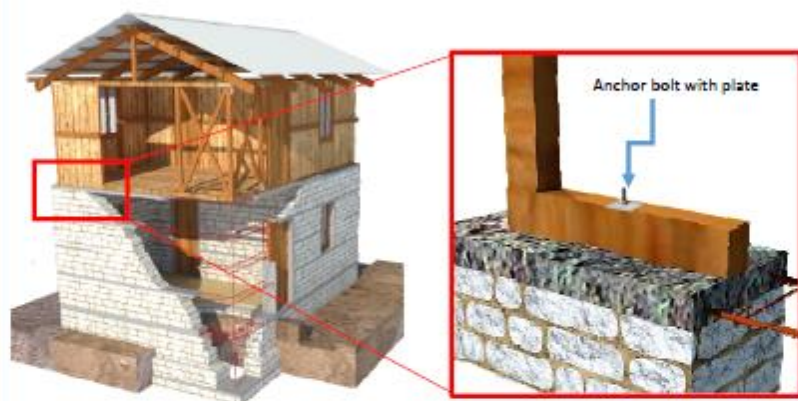
Construction methods are from

P38

Connections between RC band and wooden base plate

In case of floor band is made by **Reinforcement Concrete**.

Option 1: Connecting by anchor bolt



Anchor bolts: Minimum diameter is 12mm (M12) and length 250mm with plate 40mm x 40mm x 4.5mm.

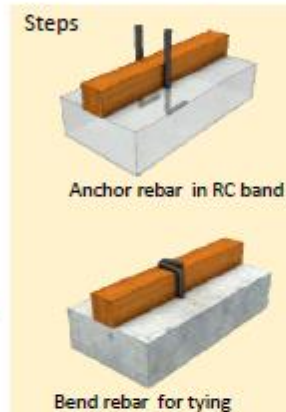
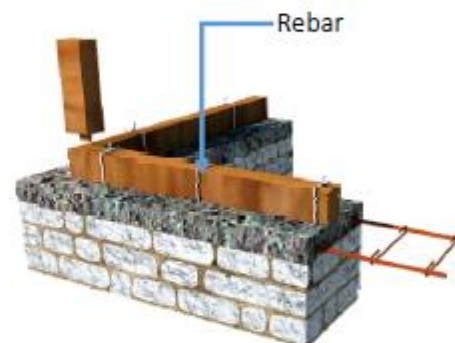
Connecting wooden base plate and RC floor bands by anchor bolts. The anchor bolt are installed at the maximum spacing is 2.0 meter.

Connections between RC band and wooden base plate

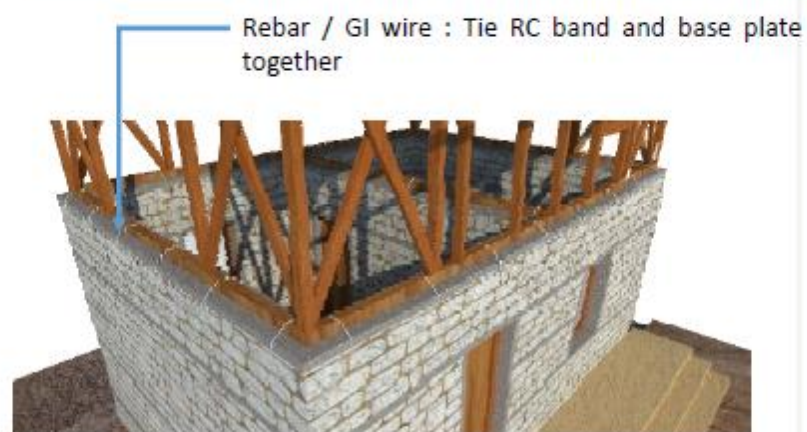
In case of floor band is made by **Reinforcement Concrete**.

Option 2: Connecting by anchoring rebar

Connecting base plate and floor bands by Rebar.



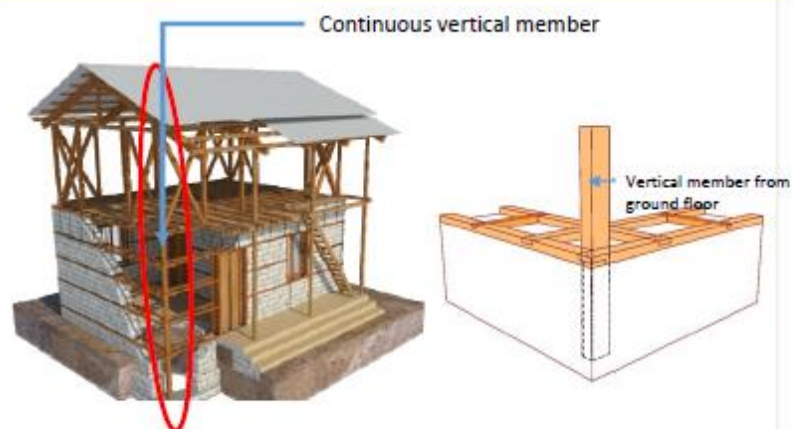
Option 3: Tying by GI wire



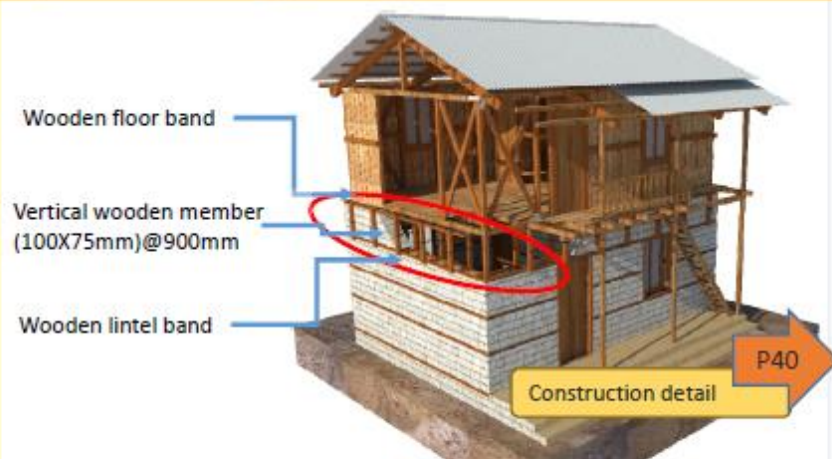
Connections between wooden band and base plate

In case of Stone masonry in **mud mortar** with **wooden band**

Option 1: Continuous vertical member

The vertical member is continuous from ground floor to 1st floor.

Option 2: Connecting roof and lintel wooden band

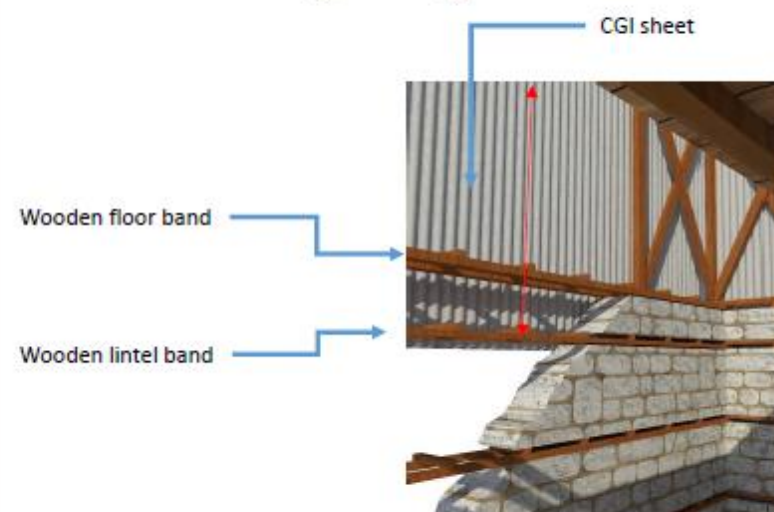


Wooden floor band and lintel band should be connected.

Connections between wooden band and base plate

In case of Stone masonry in **mud mortar** with **wooden band**

Option 3: Extending CGI sheet to lintel band



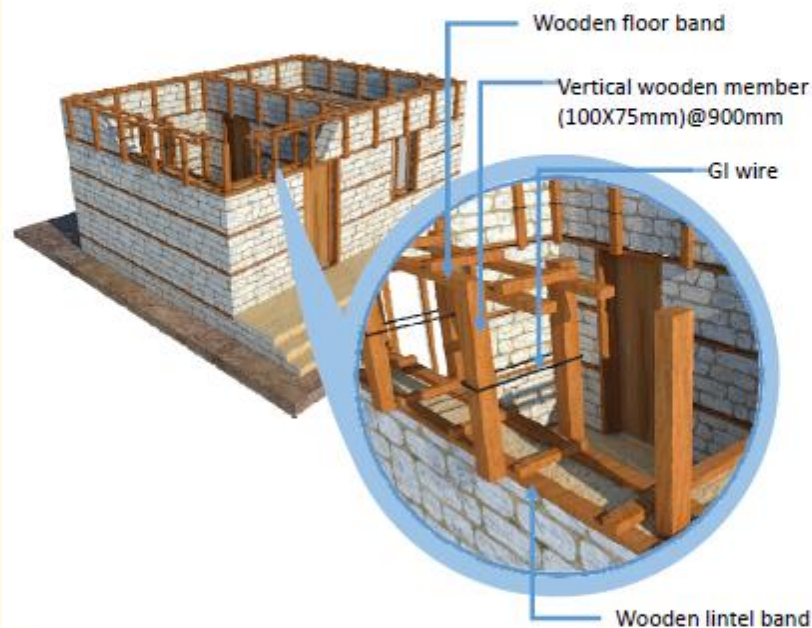
CGI sheet is extended to lintel band fixed by nail properly.

Connections between wooden band and base plate

In case of Stone masonry in **mud mortar** with **wooden band**

New construction/Correction measures

Option 2-1: Connecting wooden floor band and lintel band with vertical wooden member.



Steps:

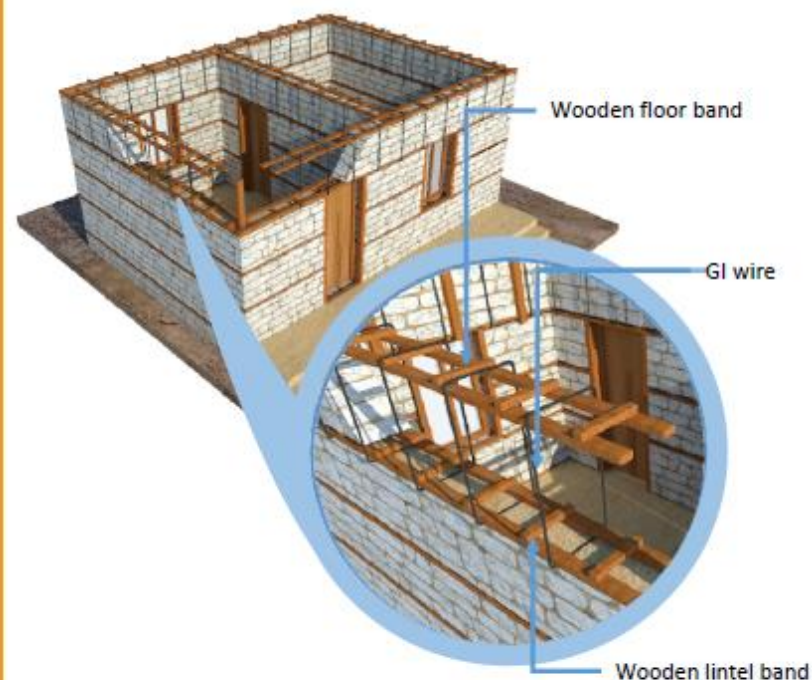
1. Connect lintel band and floor band with vertical wooden member on both side of the wall.
2. Place the vertical wooden member (100x75mm)@ 900mm c/c on both side of the wall.
3. Tie both vertical member with GI wire

Connections between wooden band and base plate

In case of Stone masonry in **mud mortar** with **wooden band**

New construction/Correction measures

Option 2-2: Connecting wooden lintel band and lintel band with GI wire

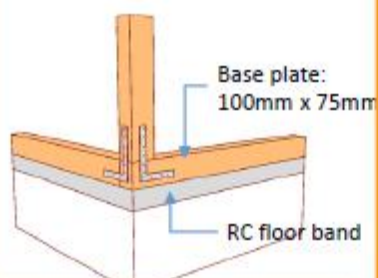


Steps:

1. Insert and tie the GI wire 3.25mm (10 Gauge) at the spacing of 450mm c/c. starting from lintel to the floor band through out the wall.

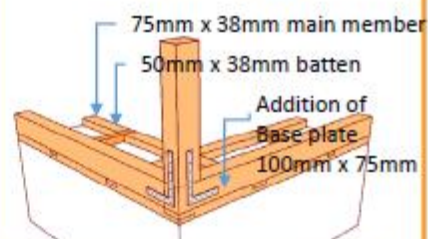
Connections between lower and upper structure

In case of RC floor band

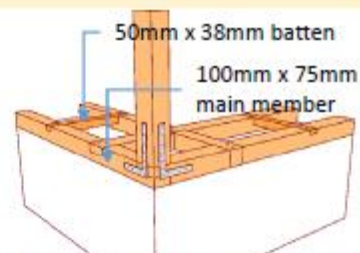


Wooden base plate should connect to RC floor band

In case of wooden floor band



If wooden band is minimum size, addition of base plate is needed.

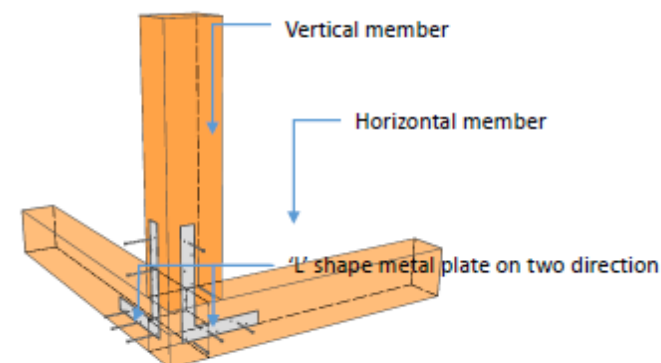


Use of larger horizontal member in floor band as per requirement of base plate

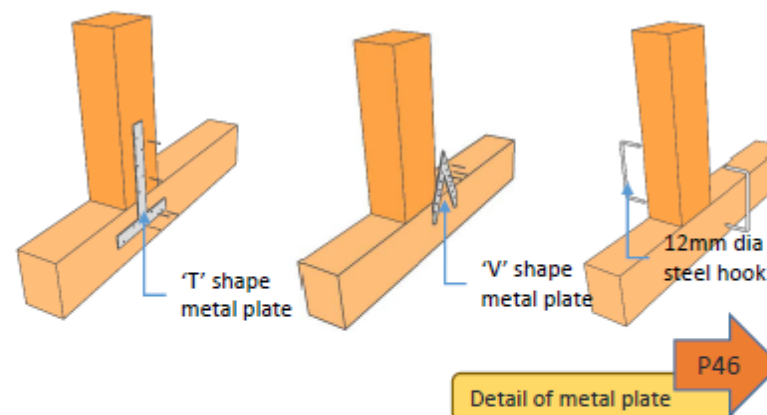
Connections between base plate and vertical member

Wooden vertical member should be properly connected to horizontal member as shown in figure.

Detail A: Connection of corner



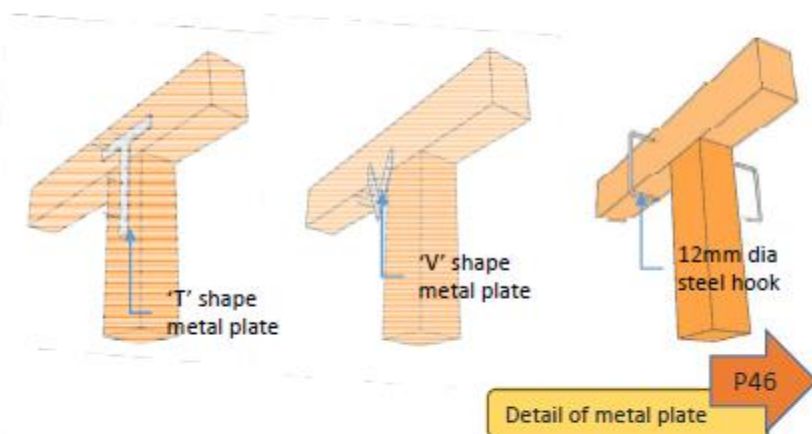
Detail B: Connection horizontal and vertical at middle



Connections between top plate, vertical and bracing member



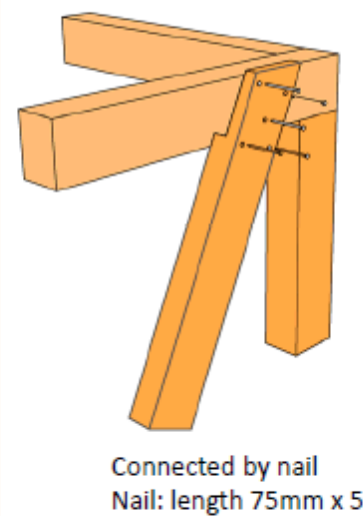
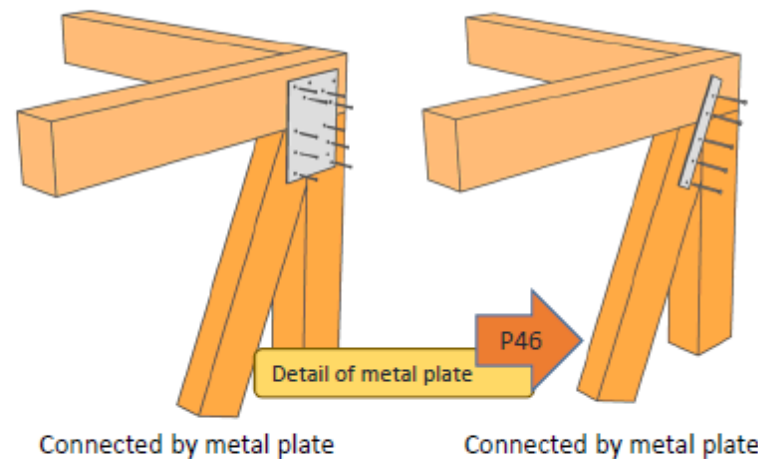
Detail B: Connection horizontal and vertical at middle



Connections between top plate, vertical and bracing member

Wooden vertical member should be properly connected to horizontal member as shown in figure.

Detail A: Connection of corner



Joint between braces



It should be fixed by nail
Nail: length 75mm x 4

Joint of timber member

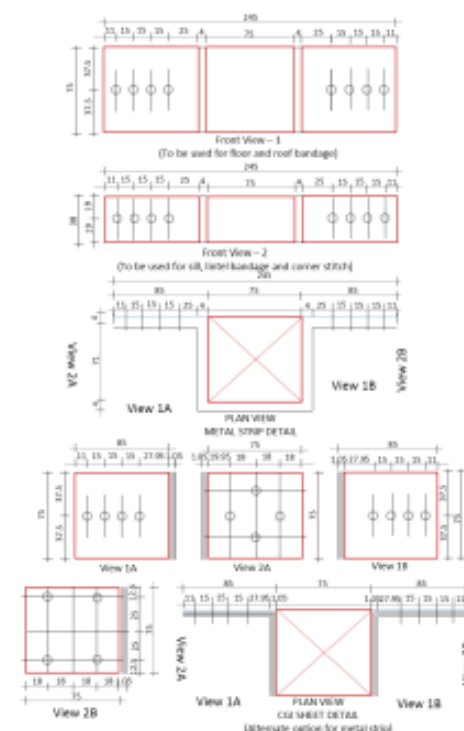
Detail of metal plates

Items	Detail	Location
T shape L-shape V-shape		Connections: between base plate and vertical member Connections: between top plate and vertical member
Metal plate 1 Metal plate 2	Note: Instead of metal plate, three layers of 0.35mm GI sheet can be used.	Connections: between horizontal, vertical and bracing member

The length of a nail shall be at least 2.5 times the thickness of the thinnest member and it shall penetrate the thicker member by 1.5 times the thickness of the thinner member, whichever is further.

Joint of timber member

Metal plate for connection between horizontal and vertical member



Inspection procedure

The detail of connection that needs to be checked are:

- Connection between wooden member.
- Connection between wall plate and floor band.
- Connection of braces with the vertical and horizontal member (base and Top plate).
- Connection between wooden lintel and floor band.

4. Frame action (Vertical, Horizontal and Bracing member)

Requirements

No.	Category	Description
4.	Frame	
		Vertical member
		It shall be continue from base plate to top plate.
		Size
		It shall be more than 75x75(mm) in hard wood, 100x75(mm) in soft wood.
		Spacing
		It shall be less than 1200mm.
	Horizontal member	It shall be continuous at same level.
		Base plate/ Top plate
	Bracing	It shall be more than 75x75(mm) in hard wood, 100x75(mm) in soft wood.
		location
		It shall be symmetrical and located at each corners.
		Direction
		It shall not be in same direction.
		Size/number
		It shall be as per the specification.

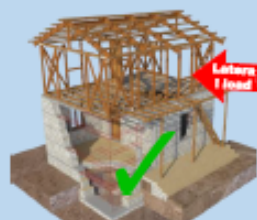
Exception

- Steel can be used instead of wood, but its strength shall be equivalent to the required strength of wood.
- If structure is found to be safe after structural calculation.
- Size of vertical and horizontal member can be variable depending upon the span. Refer to NBC 203 and 204.
- If the size of main wooden member in floor band is as per the required size of base plate, then base plate can be ignored.
- roof band is using as per requirements of base plate, base plate can be ignored.

Why important?

Earthquake-induced inertia forces will be distributed to wall consist of vertical, horizontal member and bracing. Therefore, frame should be supported horizontally and vertically. Wall framing should have diagonal braces, or sheathing boards so that the frame acts as a shear or bracing wall.

Diagonal bracing is main element to resist the frame against lateral loads due to earthquake and wind.



Technical specification

Fundamental items

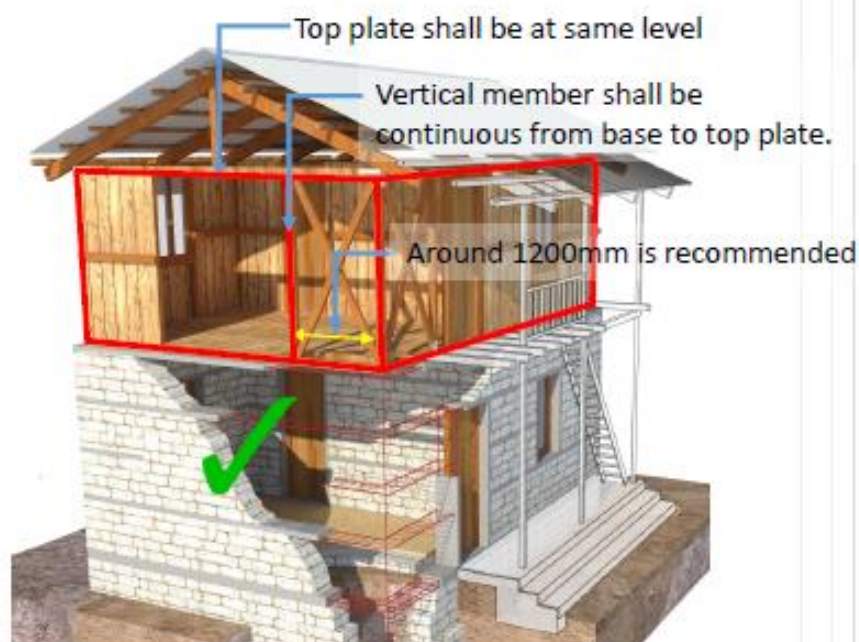
1. Vertical, horizontal member and bracing shall be properly connected.
2. Vertical member shall continue from base to top plate.
3. The spacing of vertical member shall be appropriate.
4. Horizontal member shall be continuous at same level.
5. Bracing shall be symmetrical and located at each corners.
6. Bracing shall be properly connected and provided from base to top plate.

Correction measures



Vertical and Horizontal member

1. Standard type: Lower structure is minimum requirement



Discontinuous top plate at different level

Correction measures

Strengthen gap of top plate. Adding horizontal/vertical member with proper connection.

Vertical and Horizontal member

2. Traditional type: Lower structure is exceptional of minimum requirement



Table: Size of vertical member(mm) for Various Spans

SPECIES	STOREY	COLUMN SPACING			
		2m	2.5m	3m	3.5m
Hard Wood	Ground	110x110	110x110	120x120	130x130
	1st	90x90	90x90	100x100	100x100
Soft Wood	Ground	120x120	120x120	130x130	140x140
	1st	90x90	100x100	100x100	110x110

Table: Depth of Beams (mm) for Various Spans

SPECIES	SPAN				
	2m	2 to 2.5m	2.5 to 3m	3 to 3.5m	3.5 to 4m
Hard Wood	190	220	240	270	300
Soft Wood	230	270	310	340	370

*The minimum dimensions of vertical members for different span shall be as tabulated in Table.

For hybrid structure, if span of vertical member is less than 1200mm, its size is 75x75(mm) in hard wood, 100x75(mm) in soft wood.

Technical specification

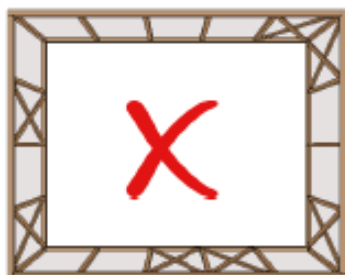
Bracing member

Location

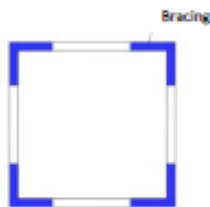
Diagonal bracing shall be located at each corner. In case of unbalanced bracing the center of gravity will be shifted and the structure will be subjected to torsion.



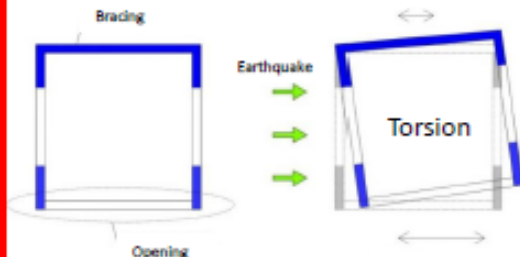
Balanced



Un-Balanced



Balanced



Un-Balanced

Technical specification

Bracing member

Direction

It shall not be in same direction.

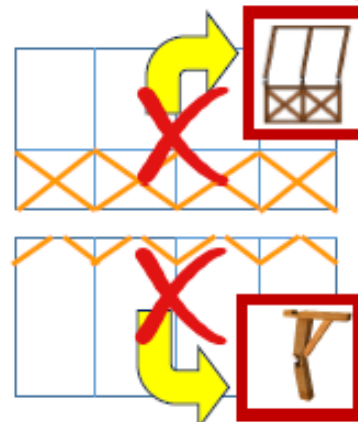
To achieve the adequate seismic resistance, provide diagonal bracing members in the planes of walls starting from base to top plate as shown in fig.



Balanced bracing



Same direction of bracing.
It shall be balanced.



Bracing shall be continuous base
plate top plate.

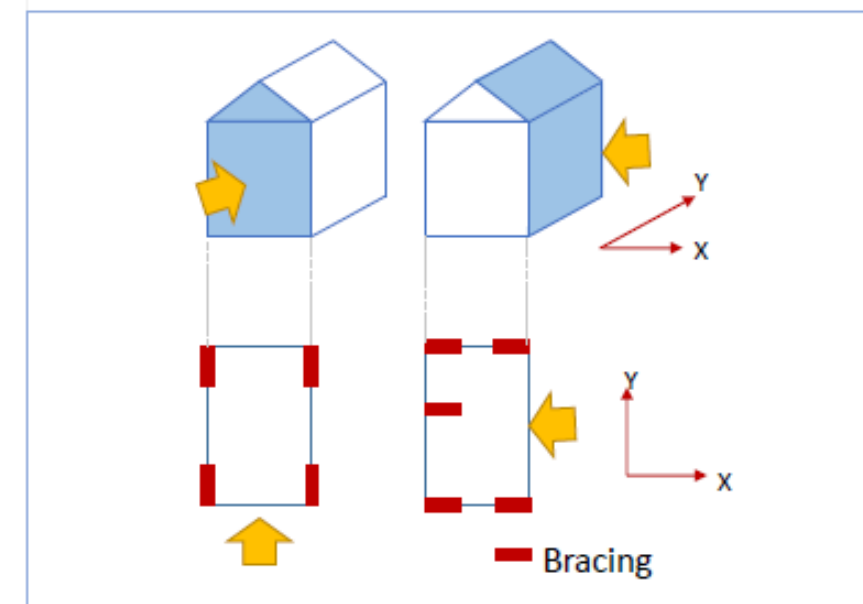
Technical specification

Bracing member

Size and Number of bracing member

Diagonal bracing is main element to resist the frame against lateral loads due to earthquake and wind.

Size and number of bracing should be consider at each X and Y direction.



Technical specification

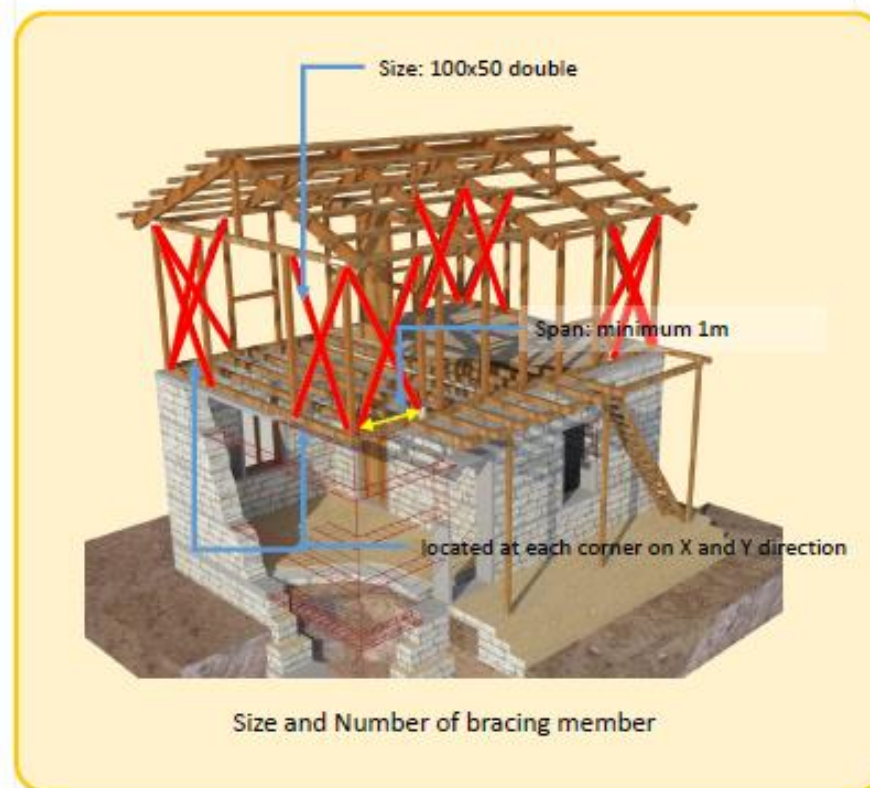
Bracing member

Size and Number of bracing member

Specification base. Inspection shall be as per the specification.

Under the following condition, inspection on the basis of specification is enabled.

- ✓ Area of building is less than 50 sq m.
- ✓ Upto 2 storey without attic.
- ✓ Wall height of first floor is less than 2.5m
- ✓ Using light weight material for roof and wall.
- ✓ And all other requirements of each item are fulfilled.



Bracing member

Size and Number of bracing member

Specification base. Inspection shall be as per the specification.

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- ✓ Using light weight material for roof and wall.
- ✓ And all other requirements of each item are fulfilled.



Specification of size and number of brace

Wooden Brace fixed by nail	Size	100 x 50 mm	double
	length	Minimum: 1meter	
	Number of each direction (X and Y)	4 (Located at each corner)	
Calculation	100x50: unit strength 2.6kN/m 2.6 x 2 (double) x 1 (meter) x 4 = 20.8kN		

If the materials and size of the bracing members vary then the simplified calculation shall be done using the shear strength provided in next page.

Bracing member

Size and Number of bracing member

Method of Bracing/ wall construction				Shear Strength of Unit wall (kN/m)
	No brace			0.0
	Mud wall	Thickness less than 50mm		1.5
		Thickness 50mm~70mm		1.8
		Thickness 70mm~90mm		2.2
		Thickness more than 90mm		2.5
	Brace rebar $\Phi 9$			1.6
 Single brace Double brace	Wooden Brace	90mm*15mm	Nail	1.6
		90mm*30mm	Steel Plate	2.4
			Nail	1.9
		90mm*45mm	Steel Plate	3.2
			Nail	2.6
90mm*90mm	Steel Plate	4.8		
	Wooden plank wall			0.8
	Structural Plywood	12mm	5.2	
	Gypsum Board	9mm	1.1	
	Plywood	3mm	0.9	

Fundamental case: Wooden brace 100x50(double) fixed by nail:

$$2.6\text{kN/m}(\text{unit strength}) \times 2 (\text{double}) \times 1.0(\text{meter}) \times 4 = 20.8\text{kN}$$

Option 1: Wooden brace 100 x 100(single) fixed by steel plate:

$$4.8\text{kN/m}(\text{unit strength}) \times 1 (\text{single}) \times 1.2(\text{meter}) \times 4 = 23.04\text{kN} > 20.8\text{kN} \quad \text{OK}$$

Option 2: Brace rebar $\Phi 9$ (double and 2pieces)

$$1.6\text{kN/m}(\text{unit strength}) \times 2(\text{pieces}) \times 2(\text{double}) \times 1.0(\text{meter}) \times 4 = 25.6\text{kN} > 20.8\text{kN} \quad \text{OK}$$

Bracing member

Size and Number of bracing member

Simplified calculation of bracing member

CONCEPT

$$\frac{Q_u}{V_u} \geq 1.0$$

Ultimate lateral strength of house shall be larger than required seismic load from code

Vu: Calculation of SEISMIC LOAD

Required Seismic force following NBC105

$$V = C_d \cdot W_t$$

seismic coefficient
 $C_d = C \cdot Z \cdot I \cdot K$

- Dead load
- Live load
- Wind load
- Snow load

<

Qu: Ultimate lateral strength of house

- Wall ratio of each direction (Ground floor and First floor)

Required seismic load from NBC105

10.1 Horizontal Seismic Base Shear

10.1.1 The horizontal seismic shear force acting at the base of the structure, in the direction being considered, shall be :

$$V = C_d W_t \quad 10.1$$

where C_d is as defined in 8.1.1.

8.1 Design Spectra and Lateral Force Coefficients

8.1.1 Design Horizontal Seismic Coefficient for the Seismic Coefficient Method

The design horizontal seismic force coefficient, C_d shall be taken as :

$$C_d = CZIK \quad 8.1$$

Where C is the basic seismic coefficient for the fundamental translational period in the direction under consideration.

Bracing member

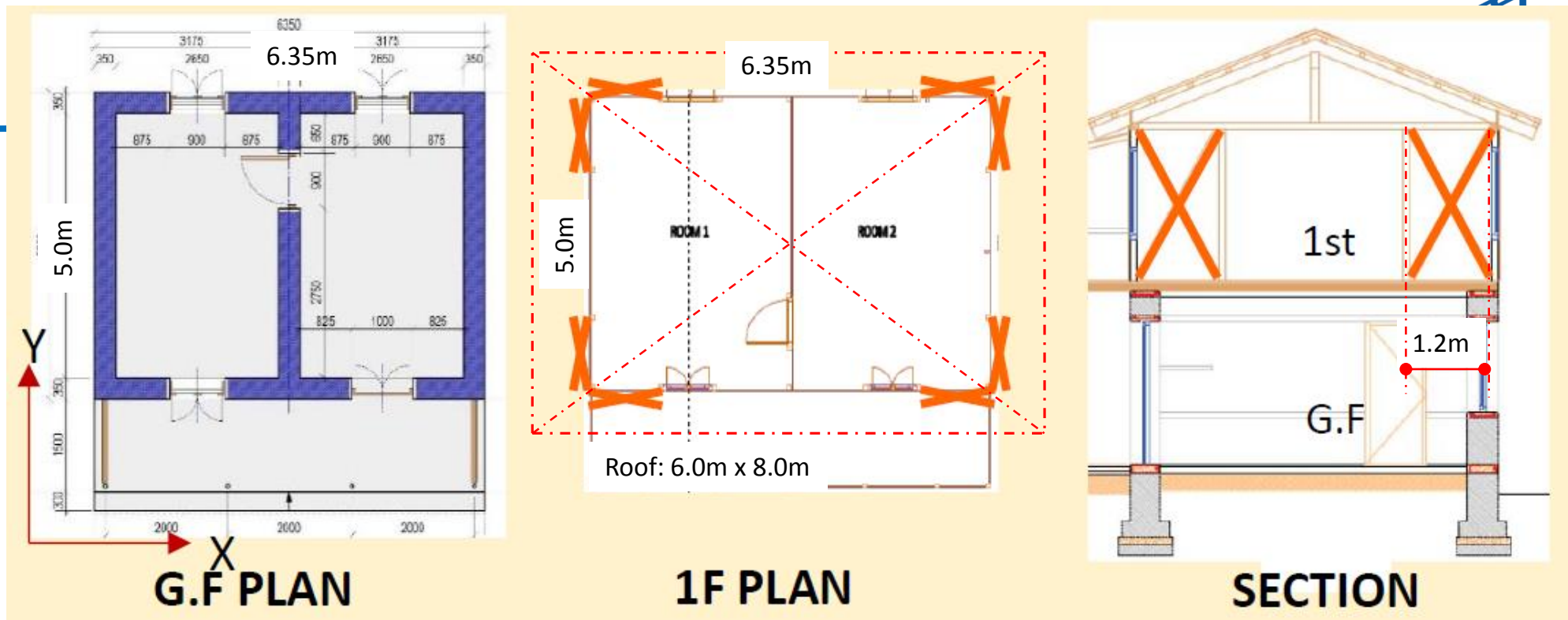
Size and Number of bracing member

Calculation base

SEISMIC LOAD	Seismic coefficient	C:	Basic seismic coefficient	①	0.08	
		Z:	Zone factor	②	1	
		I:	Importance factor	③	1	
		K:	Structural performance factor	④	4 masonry structure 2.5 frame structure	
		Cd = CZIK		⑤	0.32 masonry structure 0.2 frame structure	
	Weight of building (only 1st floor)	Roof	Unit weight	Heavy	⑥	2.52 kN/sqm
			Area	Light	⑦	0.79 kN/sqm
			Sub total	⑧	⑥x⑦	sqm
		Wall	Unit weight	Heavy	⑨	2.52 kN/sqm
			Area	Light	⑩	0.5 kN/sqm
			total length	⑪	⑨x⑩	m
			Sub total	⑫	⑨x⑩x⑪	m
		Floor (if attic is there)	Unit weight	Heavy	⑬	2.52 kN/sqm
			Area	Light	⑭	0.5 kN/sqm
			Sub total	⑮	⑬x⑭	sqm
		TOTAL WT.		⑯	⑧+⑫+⑮	kN
		Seismic load = Cd x WT		⑰	⑤x⑯	kN

Allowable strength	X-direction	brace	Type	refer from table1	①	kN/m
			single	②	1	
			double	③	2	
		length	Number	④		m
			Total length	⑤	③x④	m
		Total strength		⑥	①x②x⑤	kN
	Y-direction	brace	Type	refer from table1	⑦	kN/m
			single	⑧	1	
			double	⑨	2	
		length	Number	⑩		m
			Total length	⑪	⑨x⑩	m
		Total strength		⑫	⑦x⑧x⑪	kN

Allowable strength of each direction ⑥ and ⑫ should be bigger than seismic load ⑰

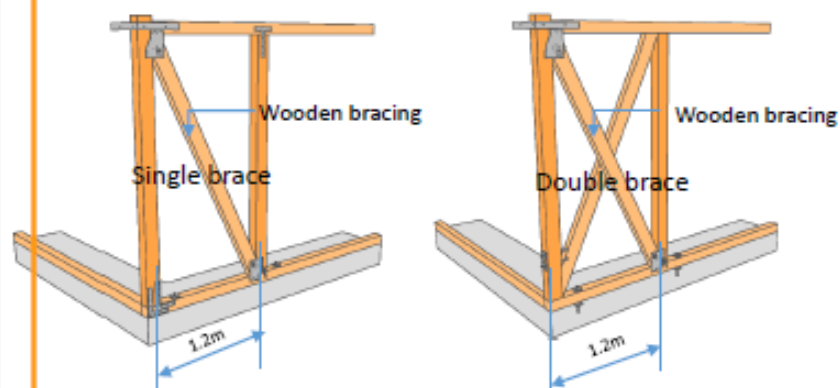


Floor area	5.0m x 6.35m	=	31.75 m ²
Roof area	6.0m x 8.0m	=	48.0 m ²
Wall area	(5.0m x 3 + 6.35m x 2) x 2.4	=	66.48 m ²

Bracing member

Size and Number of bracing member

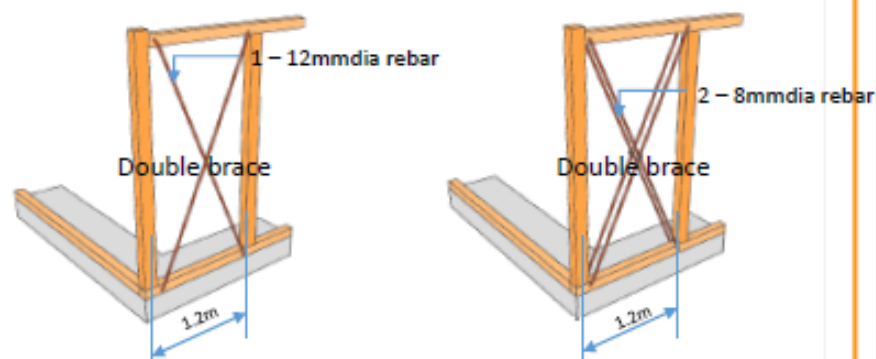
Wooden bracing member



If wooden brace 100 x 100(single) fixed by metal plate:
Shear strength:
 $4.8\text{ kN/m}(\text{unit strength}) \times 1(\text{single}) \times 1.2(\text{meter}) = 5.76\text{ kN}$

If wooden brace 100 x 50(double) fixed by metal plate:
Shear strength:
 $3.2\text{ kN/m}(\text{unit strength}) \times 2(\text{double}) \times 1.2(\text{meter}) = 7.68\text{ kN}$

Rebar bracing member



If 1-12mm dia. Rebar:
Shear strength:
 $1.6\text{ kN/m}(\text{unit strength}) \times 2(\text{double}) \times 1.2(\text{meter}) = 3.84\text{ kN}$

Instead of 12mm dia. 2 number of 8mm dia. also can be used.

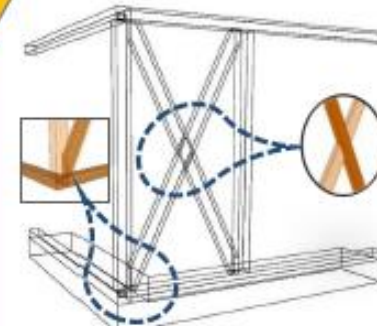
Bracing member

Size and Number of bracing member

Correction measures



No bracing member.



Steps

1. Add diagonal bracing
2. Connect vertical, horizontal member and bracing by proper connection materials such as metal plate.

5. Roof

Requirements

No	Category	Description		
7	Roof	Material	Use of light roof	
		Connection	All member shall be properly connected.	
		Bracing	For flexible diaphragm, diagonal bracing shall be considered.	

Why important?

- If heavy weight is on the top of building it will be subjected to larger seismic force. Therefore, Light weight roof is required.
- The joints of wooden roof trusses need to be bolted together and tied with metal straps as it will provides flexibility and prevent from collapse.
- In order to resist lateral forces, depending upon the structures of roof, it might be need cross bracing at all levels. It provides strength against lateral forces so that the building does not collapse sideways but is held together.

Exception

- If structure is found to be safe after structural calculation.

Inspection methodology

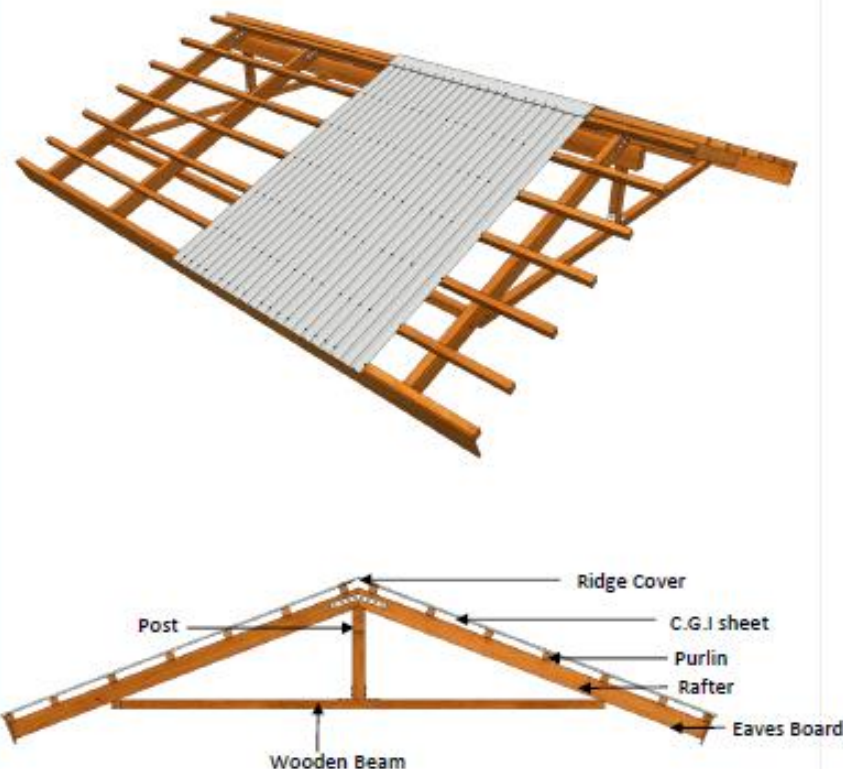
- The size of the main wooden member, batten can be identified by measurement.
- The spacing of the batten can also checked by the measurement whereas the connection can be checked by the observation.

Technical specification

Fundamental items

1. Use a continuous wall plate, ridge and purlins to tie the rafters or trusses together.
2. Stiffening of roof
 - Diagonal straps with steel nut bolts or metal nails
 - Diagonal steel truss with steel nut bolts or metal nails
 - Timber bracing with metal nails or timber nails

Wooden Roof truss



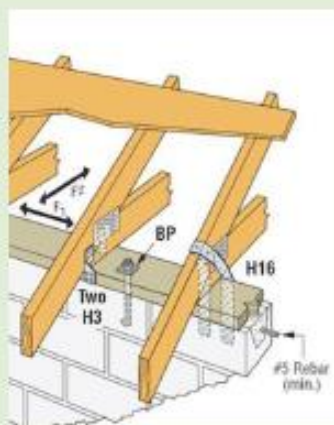
Strengthening roof

Connection improvement between wall to roof

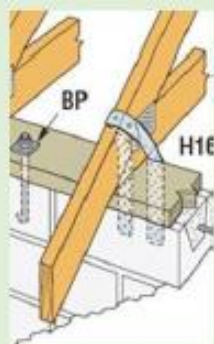
Correction measures

Option : from Retrofitting manual.

- Metal Strap with Screws



Note : 3 mm thick metal strap, Minimum four numbers of 50 mm long nails (Fe250) with Floor member and Minimum four numbers of M16 grade expansion bolts with walling material



Details of Anchor plate



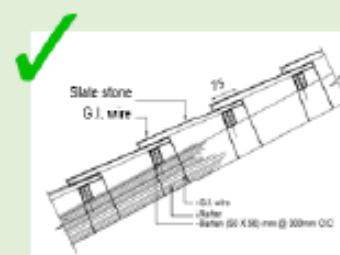
Strengthening roof

Sliding of Roofing materials

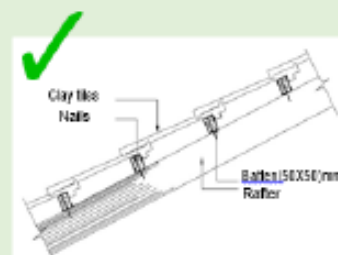
Correction measures

Option : from Retrofitting manual. Fixing roofing tile.

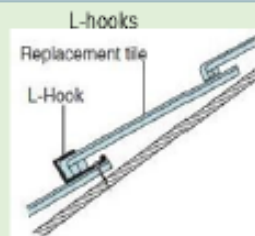
1. Replace damaged tiles.
2. Using appropriate correct fixing method for roofing materials.
3. Connect the roof with the roof band by inserting reinforcement or GI sheet.
4. Slate stone and clay tiles should be properly anchored to purlin as NBC.



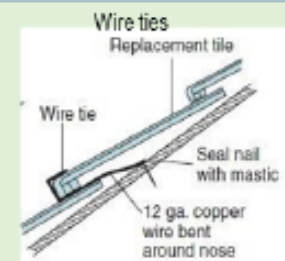
Detail of anchoring slate stone
Source: NBC203:2015, P44



Detail of fixing clay tile
Source: NBC203:2015, P45



Gently lift the overlapping tile and twist loose the damaged tile.
After filling any nail holes, slip in a new tile and secure with an L-hooks (left) or bent copper wire (right)



Inspection sheet

Final Inspection of Hybrid Structure



Government of Nepal
Ministry of Urban Development
Central Level Project Implementation Unit

INSPECTION SHEET OF HYBRID STRUCTURE FOR FINAL INSPECTION									
Information of House Owner/Beneficiary				Date of Inspection		dd-mm-yyyy			
Name:				Grand Agreement No.					
Address:				VDC/Municipality		ward		tole	
								Land plot No	
SECTION -1: DESCRIPTION PROVIDED IN THE APPLICATION TO SURVEY THE HOUSE									
If use fix design from design catalogue,				Design No.					
If free design by house owner				Technique and Construction material					
Fill construction typology from P.A form				Construction of roof and materials					
Technical Assistant				☐ YES, ☐ NO		Organization		☐ GoN, ☐ NGO ()	
Trained Masons used				☐ YES, ☐ NO		Soil type		☐ Hard, ☐ Medium, ☐ Soft	
SECTION-2: DETAIL TECHNICAL INSPECTION									
S.N	Category	Description		Comply		Remarks			
				YES	NO				
1.	Shape of House	No. of storey	Not more than two storey	☐	☐				
		Shape of house/Span of wall	1 st floor Regular shape. The wall line of upper storey shall be on the wall of lower storey. The wall line shall not be cantilevered. Therefore, the span of wall also same as lower storey.	☐	☐				
		Height of wall	1 st floor It shall not be more than 2.5m.	☐	☐				
2.	Materials	Nail	Common wire nails shall be made of mild steel having a minimum tensile strength of 550N/mm ² . Nails with appropriate diameter and length shall be provided.	☐	☐				
		Bolt	The number, diameter, length and spacing shall be as per the specification. (*refer hybrid manual)	☐	☐				
		Metal plate	The number, diameter, length and spacing shall be as per the specification. (*refer hybrid manual)	☐	☐				
		Rebar	High strength deformed bars with fy = 415 Mpa /500 Mpa.	☐	☐				
		Timber	Treated and well seasoned hard wood or locally available wood without knots shall be used.	☐	☐				
3.	Connection and Joints	Connection between lower and upper structure	It shall be properly connected as per the specification. (*refer hybrid manual) In SMM or BMM with timber band, the floor band and lintel band shall be properly connected.	☐	☐				
		Joints of structural member	All the structural members shall be properly connected by nails, bolts and metal plate as per the specification. (*refer hybrid manual)	☐	☐				
4.	Frame	Vertical member	It shall be continue from base plate to top plate with proper connection.	☐	☐				
		Size	It shall not be less than 75X75(mm) for hard wood and 100 X 75(mm) for soft wood.	☐	☐				
		Spacing	It shall be less than 1200mm.	☐	☐				
		Horizontal member	It shall be continuous at same level.	☐	☐				
		Base plate/Top plate	It shall be (100 X 75)mm. It shall be properly connected with floor band. It shall not be less than 75X 75 mm in hard wood and 100 X 75(mm) in soft wood.	☐	☐				
Bracing member	Location	It shall be symmetrical and located at each corners.	☐	☐					

		Direction	It shall not be in same direction.	☐	☐	
		Size/number	It shall be as per the specification (*refer hybrid manual)	☐	☐	
5	Roof	Material	Use light roof	☐	☐	
		Connection	All member connected properly	☐	☐	
		Bracing	For flexible diaphragm, Diagonal bracing shall be considered.	☐	☐	

Others:

- At least four number of photographs with their number
- Tentative drawings of building:
- After the detail description of the under constructed house, is it satisfactory to give completion certificate

Yes ☐ , No ☐

☐ It was passed through the inspection of the third inspection so the construction can be provided building completion certificate from VDC/Municipality.

☐ If was found to be corrected/retrofitted so correction order is given using Annex-6

- Acceptation of Description provided agreeing that the technical details during inspection is correct:

House owner/Beneficiaries or representative name:.....Signature:.....

Relationship with house owner (In case of representative):.....Date:.....

- Submit for Approval of the technical inspection:.....

MOUD-DLPIU Supervisor:.....

Name..... Designation:.....

Signature..... Date.....

- Approved by:

MOUD DLPIU Supervision Engineer.....

Designation:.....

Signature..... Date.....

Thank You